DEMANDIQ · PEAK-SEASON PLANNING BRIEF

Plan the outerwear peak on demand, not on last year's sales

Premium outerwear demand planning case study · Abhijit Mishra · retrospective decision evaluated at the June 29, 2026 planning date · simulated data

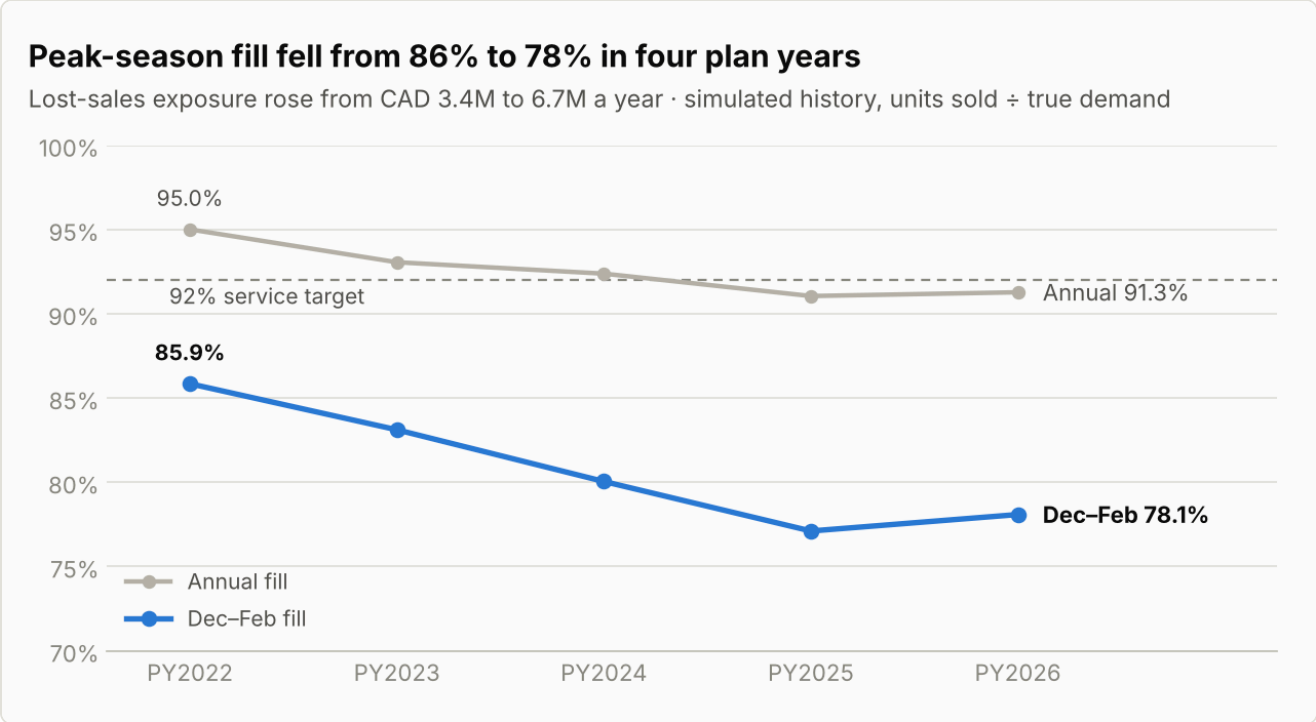
Peak-season fill fell from 86% to 78% in four plan years because every buy starts from last year's sales, and stockouts suppress exactly the peak weeks. Evaluated at the June 29, 2026 planning date, planning the coming peak on reconstructed demand meant an indicative **10,339 more units (+22% on last peak's Dec–Feb sales)**, with orders due by **Jun 29–Jul 13**. Those deadlines have passed; this brief shows the decision the planner should have made then.

78.1%
Dec–Feb fill in PY2026, down from 85.9% in PY2022 *(truth)*

CAD 6.65M
lost revenue, 9.1% of net revenue *(truth, upper bound)*

+10,339 units
indicative peak change at the planning date: 4,574 moved, 5,613 new *(observable)*

Jun 29–Jul 13
22-week order deadlines at the planning date, for arrival before the first stockout (now past)



Why it happens

- **The buy plans the business down.** Each buy starts from last year's sales, 9.0% below last year's reconstructed demand. In Dec–Feb, reconstructed demand runs 21% above sales. observable
- **Most of the loss is timing.** 59% of lost units had stock arriving within the plan year, but after the stockouts. truth
- **Chase is aimed at the wrong signal.** It fires in mid-September in 75–77% of cells whatever their eventual loss, and 5,403 units of chase capacity went unused. observable
- **The published 13-week plan looked at the wrong season.** Its summer window held 1.1% of lost demand, and its 3 escalations assume zero replenishment. Weather changes none of the 9 planner actions; adding estimated replenishment changes 6. observable

Evidence labels: **observable** = sales, in-stock days, receipts, inventory and reconstructed demand, which a planner can see. **Truth** = hidden simulation fields, used only to size the problem and to validate.

What the planner should have decided on June 29, 2026

Decision	Quantity	Deadline · owner
Plan the peak on reconstructed demand; move stock before buying	Indicative +10,339: 4,574 moved forward from spring replenishment, 5,613 new, 152 accepted gap (excess guardrail)	Commit suppliers by Jun 29 (6 series) or Jul 13 (3) · planner, S&OP, supply
Size chase from the peak gap, not early shipments	Fallback floor of about 4,355 units within last season's unused chase capacity	Chase order by Sep 21 or Oct 5 · planner with supply planning
Rank exceptions by stockout days, by SKU × region × channel	Top 20 of 81 cells by stockout days capture 53.4% of true lost revenue (true top 20: 60.2%)	Each monthly S&OP · Demand Planner
Escalate only shortfalls that survive forecast error	If 6,512 estimated replenishment units arrive as modelled (by Jul 13; unconfirmed): 0 escalate, 7 protect, 2 hold; error band adds 1 watch. Zero replenishment (published): 3 escalate.	Weekly S&OE · planner confirms open orders and receipt dates before replacing the published list
Release launch reserve on a 5-pp channel-mix rule	991-unit reserve on a 8,259-unit buy; ECOM lost 159 units while it sat idle	Launch checkpoints W1–W2

Peak scenario by series observable

Series	Gap vs sales	Change	Moved	New	Call
IMH-001 ECOM	+20%	2,355	1,308	1,047	Buy and re-time
APS-001 ECOM	+21%	960	593	367	Buy and re-time
APS-001 Retail	+32%	958	343	615	Buy and re-time
IMH-001 Retail	+18%	1,599	674	925	Buy and re-time
IMH-001 Wholesale	+46%	2,492	927	1,565	Buy and re-time
CTS-001 Retail	+15%	625	161	464	Buy and re-time
APS-001 Wholesale	+27%	690	212	478	Buy and re-time
CTS-001 Wholesale	+11%	341	189	152	Buy and re-time
CTS-001 ECOM	+5%	319	167	0	Re-time only
Total		**10,339**	**4,574**	**5,613**	**152 units accepted as a gap**

Change = (reconstructed − sold) Dec–Feb units × (1 + SKU growth), allocated to regions by stockout days. Unsold spring-replenishment stock moves first; the rest is new. No new units where over-buy exposure is at least 50% of estimated lost revenue (policy).

Could it have landed in time? (at the planning date; all dates now past)

- **New units:** the regular October-drop order date (Apr 27) had already passed. New 22-week orders had to be placed by Jun 29–Jul 13; supplier capacity was **unconfirmed**.
- **Moved units:** they had to ship 17–21 weeks earlier than last season's spring receipts, which needed a supplier order change. Order-change windows are not modelled: **unconfirmed**.
- **Fallback:** chase alone could have landed about 4,355 units, with orders due by Sep 21–Oct 5. New units fit last season's chase headroom in 3 of 8 series.

How sure are we?

- **Direction: high.** Scored against simulation truth, the priority ranking matches 9 of 9 and every call survives recomputation with true demand. The observable top 20 cells capture 53.4% of true lost revenue, against 60.2% for the true top 20. truth
- **Size: conservative.** Reconstruction finds 75% of the true Dec–Feb gap. The planner view also asks for more new units than truth suggests (54% vs 41%). truth
- **Weekly escalations: conditional.** The replenishment estimate missed by 57–71% series-week in backtests. One published escalation needs 30% of its estimate to clear; a 29.5% cut is one scenario, not a bound.
- **Limits.** Simulated data with the failure modes designed in; the test is that the diagnosis finds them. Lost revenue assumes no substitution; supplier capacity, MOQs and order-change windows are not modelled.

Every number is generated from the CSVs: **python 04_scripts/run_pipeline.py --check** · github.com/Abhijit2409/demandiq-retail-demand-planning · simulated data